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Embedding beyond electrostatics—The role of wave function confinement

Lina J. Nåbo,^{1,a)} Jógvan Magnus Haugaard Olsen,¹ Nanna Holmgaard List,²
 Lukasz M. Solanko,³ Daniel Wüstner,⁴ and Jacob Kongsted^{1,b)}

¹*Department of Physics, Chemistry and Pharmacy, University of Southern Denmark, Campusvej 55, 5230 Odense M, Denmark*

²*Division of Theoretical Chemistry and Biology, School of Biotechnology, KTH Royal Institute of Technology, SE-106 91 Stockholm, Sweden*

³*Orphazyme ApS, Ole Maaløes Vej 3, 2200 Copenhagen N, Denmark*

⁴*Department of Biochemistry and Molecular Biology, University of Southern Denmark, Campusvej 55, 5230 Odense M, Denmark*

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We study excited states of cholesterol in solution and show that, in this specific case, solute wave-function confinement is the main effect of the solvent. This is rationalized on the basis of the polarizable density embedding scheme, which in addition to polarizable embedding includes non-electrostatic repulsion that effectively confines the solute wave function to its cavity. We illustrate how the inclusion of non-electrostatic repulsion results in a successful identification of the intense $\pi \rightarrow \pi^*$ transition, which was not possible using an embedding method that only includes electrostatics. This underlines the importance of non-electrostatic repulsion in quantum-mechanical embedding-based methods. *Published by AIP Publishing.* [<http://dx.doi.org/10.1063/1.4962367>]

I. INTRODUCTION

Theoretical predictions of solvent effects on excitation energies, i.e., solvatochromism, have for a long time been a central challenge to quantum chemistry.¹ Today it is possible to routinely perform accurate quantum chemical calculations of molecular properties for small and medium-sized molecules, but the same cannot be said for molecules subjected to an environment, as, for example, a solute in a solvent. Even if linearly scaling electronic-structure methods based on density functional theory (DFT) can be applied to very large and complex systems, there are still problems with current exchange-correlation functionals, which, for example, can result in incorrect predictions of partial electron transfer in extended systems.^{2,3} Such errors affect the electrostatic potential and may thus give rise to poorly described solvent effects. Indeed, the conventional view of solvent effects is that the solvent molecules electrostatically perturb the electronic structure of the solute and thereby change the energetics of the ground and excited states associated with the solute molecule. For this reason, effective solvent models are usually constructed by only considering electrostatic effects in the effective Hamiltonian for the solute. For instance, this is the case for most dielectric continuum models,⁴ as well as models building on the concepts of combined quantum mechanics and molecular mechanics (QM/MM).^{5–8} In such effective electrostatics-only schemes, the explicit electronic structures of the solvent molecules are neglected, meaning that the requirement of a total solute-solvent anti-symmetric wave function is discarded. As a consequence, the solute

wave function may artificially extend into regions that would otherwise be occupied by the solvent charge densities, the so-called electron-spill effect. For localized valence transitions, this is usually not a problem, but in cases where the solute excited states are rather diffuse, this can naturally cause a violation of the Pauli exclusion principle and thereby lead to unphysical results.⁹ Hence, in such cases confinement of the solute wave function cannot be neglected and should be accounted for as an operator in the effective Hamiltonian.

Here we study solvent effects on the excited states of cholesterol and show that, in this specific case, solute wave-function confinement plays a major role and is in fact crucial for even a qualitatively correct description of the electronic absorption spectrum. In general, not much is known about the impact of wave-function confinement on transition and excited-state properties of solutes in solvents,^{10–17} and the aim of this work is to demonstrate how such knowledge may be obtained as well as to underline the significance of these confinement effects. To this end, we show that the recently presented polarizable density embedding (PDE)¹⁸ approach, which is an extension of the polarizable embedding (PE)^{19,20} model, can successfully be used to study difficult cases like the absorption spectrum of cholesterol and can thus generally and very effectively be used to provide new physical insight into the old question of solvatochromism.

II. COMPUTATIONAL METHODOLOGY

In both the PE and PDE models, a total molecular system is divided into two sub-systems consisting of a core region and an outer region. The core region defines the part of the system treated using quantum mechanics while the outer region consists of solvent molecules and is

^{a)}E-mail: naabo@sdu.dk

^{b)}E-mail: kongsted@sdu.dk

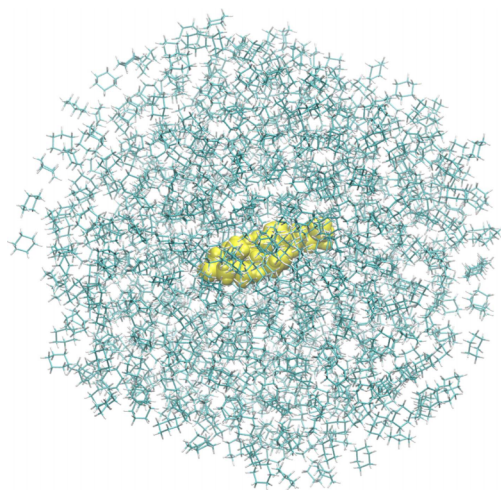


FIG. 1. A snapshot from the MD simulation of a cholesterol molecule (yellow) solvated in 692 cyclohexane molecules. The figure was produced with the Visual Molecular Dynamics program (VMD).²¹

described effectively. Figure 1 shows the molecular system studied here, where cholesterol (yellow) constitutes the core region and the cyclohexane molecules belong to the outer region. The permanent charge distributions of the fragments (molecules) in the outer region are represented either by electronic densities (together with the nuclei) as in the PDE model or by distributed electric multipole moments as in the PE model. Furthermore, polarization between all fragments, as well as the core region, results in induced charge distributions that are described at the level of induced dipole moments obtained from distributed anisotropic polarizabilities. Finally, within the PDE model, non-electrostatic repulsion between the wave functions of the core and the outer region is modeled by assigning a repulsive potential to each of the fragments in the outer region. The effective Kohn-Sham (KS) operator (or Fock operator in the case of Hartree-Fock (HF) theory) for the core region thus takes the form

$$\hat{f}^{\text{eff}} = \hat{f}^{\text{core}} + \hat{v}^{\text{es}} + \hat{v}^{\text{ind}} + \hat{v}^{\text{rep}}, \quad (1)$$

where $\hat{f}^{\text{core}}$ is the usual vacuum KS (or Fock) operator while the last three terms define the effective embedding operator $\hat{v}^{\text{PDE}} = \hat{v}^{\text{es}} + \hat{v}^{\text{ind}} + \hat{v}^{\text{rep}}$, consisting of electrostatics, induction (polarization), and non-electrostatic repulsion, respectively. The explicit expressions for the operators entering the embedding potential can be found in Ref. 18, and here we will focus only on the contribution from the repulsive potential within the PDE model. This term models short-range, non-electrostatic repulsion effects (also called Pauli or exchange-repulsion) between the core and the fragments in the outer region. This operator is based on the projection operator method by Huzinaga and Cantu,²² and models wave-function orthogonality between the core and outer-region fragments. A KS (or Fock) matrix element of this operator is given by

$$v_{\mu\nu}^{\text{rep}} = - \sum_{a=1}^{N_{\text{out}}} \sum_{\gamma\delta \in a} W_{\gamma\delta} S_{\mu\gamma} S_{\delta\nu}, \quad (2)$$

where μ and ν are the atomic-orbital (AO) basis functions belonging to the core region, γ and δ are the AO basis

functions belonging to fragment a , and N_{out} is the total number of fragments in the outer region. The element $S_{\mu\gamma}$ is an intermolecular overlap integral, and $W_{\gamma\delta}$ is an element of the energy-weighted density matrix defined as

$$W_{\gamma\delta} = \sum_{i=1}^{N_{\text{occ}}} \varepsilon_i C_{\gamma i} C_{\delta i}. \quad (3)$$

The sum in Eq. (3) is over the occupied molecular orbitals (MOs) of the outer-region fragments. Thus, ε_i is an MO energy and $C_{\gamma i}$ is an MO coefficient. These are calculated for each fragment in isolation. As seen from these expressions, the effect of the repulsive operator is to create an energy penalty. When used in an energy optimization procedure for the core-region wave function, the energy penalty ensures a solution that limits the overlap of the core wave function with the wave functions of the fragments in the outer region. Consequently, the effect of the repulsion operator is a confinement of the solute wave function to its cavity. The calculation of excited states and properties is made possible through PE and PDE implementations within the framework of resonance-divergent quantum-mechanical response theory^{23,24} in terms of self-consistent field theories, i.e., time-dependent HF (TDHF) and time-dependent DFT (TDDFT).^{18–20} In addition, to calculate absorption properties over a frequency range with a high density of states, we use the resonance-convergent formulation of response theory (damped response theory).^{25,26} To this end, we have in this work implemented PDE within resonance-convergent TDDFT following the previously outlined strategies for resonance-convergent PE-TDDFT.²⁷

The aim of this work is twofold: (i) to investigate the role of solvent effects on electronic absorption using cholesterol solvated in cyclohexane as a case study, and (ii) to illustrate how these solvent effects can be addressed computationally. In order to sample different solvent conformations, the molecular structures used in this study were extracted from a molecular dynamics (MD) simulation of a geometrically constrained cholesterol molecule solvated in cyclohexane molecules to a distance of ~ 24 Å, giving 692 cyclohexane molecules. Since no force fields for cholesterol have been developed with the aim of producing structures to be used in electronic-structure calculations, we choose to use a rigid, quantum-mechanically geometry-optimized structure in the MD simulation. Snapshots were extracted every 250 ps from a 5 ns trajectory, rendering 20 molecular structures. A detailed description of the protocol for the MD simulation is given in Section II of the [supplementary material](#). For the PE- and PDE-DFT calculations, cholesterol constitutes the core region and all the cyclohexane molecules belong to the outer region. The permanent charge distributions of the solvent molecules are represented either directly by their charge densities (PDE) or in terms of distributed multipoles (PE). The induced charge distributions are modeled using distributed polarizabilities in both models (see Section III of the [supplementary material](#) for details on functional and basis set). All PE- and PDE-DFT calculations were performed using a development version of the Dalton program²⁸ and the Polarizable Embedding library (PElib).²⁹ Excitation energies for cholesterol in gas phase or in solution were calculated using the range-separated

CAM-B3LYP³⁰ exchange-correlation functional within the TDDFT formalism employing the aug-cc-pVDZ^{31,32} basis set, and using either PE or PDE to model the solvent. Corresponding time-dependent HF results were also computed in Dalton. Gas phase reference calculations of cholesterol using the resolution of the identity (RI) approximation in the approximate coupled cluster singles and doubles method (CC2)³³ were performed in Turbomole³⁴ with the aug-cc-pVDZ basis set. Finally, a full QM cluster calculation using TDHF was performed in LSDalton.²⁸ Calculations using resonance-convergent TDDFT employed a damping parameter (γ) of 1000 cm⁻¹.

III. RESULTS AND DISCUSSION

We first discuss experimental findings related to the electronic absorption spectrum of cholesterol. An experimental UV absorption spectrum of a solution consisting of 2.59 mM cholesterol in cyclohexane has been recorded, and one clear absorption maximum at 5.82 eV is identified. The experimental details can be found in Section I of the [supplementary material](#). Cholesterol possesses one double bond, and it would therefore be expected that the absorption is dominated by a localized $\pi \rightarrow \pi^*$ transition. In order to evaluate this hypothesis and characterize the absorption spectrum, the electronic transition properties of cholesterol were investigated computationally. The initial approach was to perform the calculations in gas phase. With compact basis sets, the lowest excitation is indeed a localized $\pi \rightarrow \pi^*$ transition (dominated by a transition from the highest occupied molecular orbital (HOMO) to the lowest unoccupied molecular orbital (LUMO)) and leads to an intense absorption, as reported in Table S1 in the [supplementary material](#). However, with the addition of diffuse basis functions to the basis set, the $\pi \rightarrow \pi^*$ transition can no longer be identified and instead, many low-lying, diffuse excited states with low or zero intensity emerge. This finding is similar across different classes of basis sets, as can also be seen in Table S1. To verify that an intense $\pi \rightarrow \pi^*$ transition is not still present at energies above the diffuse excited states, the absorption spectrum for cholesterol was evaluated using resonance-convergent TDDFT, employing the aug-cc-pVDZ basis set, a rather flexible and diffuse basis set (see black line in Fig. 2). This spectrum confirms the absence of one clear absorption, and any intensity in the UV spectrum up to at least 8.0 eV stems from transitions to multiple diffuse states. This was found by visualization of the dominating orbitals in the excitation vectors from the corresponding resonance-divergent TDDFT calculation (see Table S4 in the [supplementary material](#) for excitation energies and oscillator strengths for the ten lowest transitions). Corresponding TDHF calculations were performed to ensure that the diffuse character of the excited states is not only an artifact of the exchange-correlation functional used in the TDDFT calculations. Even though a slightly larger intensity is observed for some of the lowest transitions in the resonance-divergent TDHF calculation (see Table S3 in the [supplementary material](#)), the TDHF results show the same qualitative picture as TDDFT (see Fig. S1 in the [supplementary material](#) for the resonance-convergent

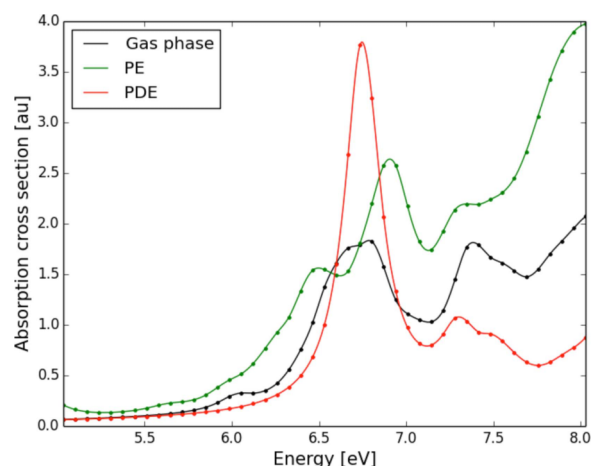


FIG. 2. Absorption spectra of cholesterol obtained using resonance-convergent TDDFT with CAM-B3LYP/aug-cc-pVDZ in gas phase (black) or solvated in cyclohexane modeled using the PE (single snapshot, green) or the PDE model (single snapshot, red). The damping parameter is 1000 cm⁻¹.

TDHF spectrum). The validity of the gas phase CAM-B3LYP calculations is further confirmed by a calculation using RI-CC2, which also shows many low-lying transitions with little or no intensity in gas phase. The agreement between the CAM-B3LYP and RI-CC2 results (see Tables S4 and S5 in the [supplementary material](#)) indicates that the diffuse character of the excited states is neither an artifact of missing dynamical correlation nor of the exchange-correlation functional. Since we consider RI-CC2 as our reference method, we assume that the intensity obtained in the TDHF calculations is present due to missing correlation effects. The gas phase calculations are further discussed in Section IV of the [supplementary material](#).

Based on these findings, it is clear that the gas phase calculations do not reproduce the experimental results in solvent when basis sets including diffuse basis functions are used. This observation leads to the assumption that solvent effects need to be taken into account in the calculations. In order to explicitly include the electrostatic effect of the solvent, the PE model was initially employed for a single snapshot extracted from the MD simulation (see Fig. 1). However, the expected intense $\pi \rightarrow \pi^*$ transition cannot be found in the relevant part of the spectrum (see green line in Fig. 2) and the spectrum is still characterized by weak transitions to many close-lying, diffuse states. The rise in intensity at high energies is likely due to an increasing density of states in this region. It appears that for an accurate description of the electronic transitions in cholesterol, the inclusion of solvent electrostatics only is not a sufficient description of the environment, as might be expected for a non-polar solvent as cyclohexane. The quantum nature of the solvent molecules is not taken into account in the PE model, where only classical electrostatics is modeled by the distributed multipoles and induced dipoles. The PDE model, however, reintroduces the quantum nature through the repulsion operator (Eq. (2)). In other words, the steric effect of the solvent electronic densities is described, which consequently restricts the volume accessible to the charge distribution of cholesterol. When the transition properties of cholesterol are calculated with the PDE model and the aug-cc-pVDZ basis set, using the same

snapshot as before, an intense $\pi \rightarrow \pi^*$ transition is clearly identified as the lowest excitation positioned at 6.75 eV and with an oscillator strength of 0.36. The absorption properties of cholesterol within the PDE model were also calculated using resonance-convergent TDDFT, and the resulting spectrum (see red line in Fig. 2) illustrates how the intense excitation emerges when the cholesterol wave function is confined by the solvent molecules, as compared to the weaker absorption resulting from transitions to multiple diffuse states seen both in gas phase and in solution when the solvent is modeled using the PE model. The cholesterol absorption spectra discussed here were also evaluated using resonance-convergent TDHF. As shown in Fig. S1 (supplementary material), PE- and PDE-TDHF qualitatively reproduce the solvent effects predicted by the PE- and PDE-TDDFT calculations.

Based on the PDE-TDDFT calculations, it is seen that cholesterol-cyclohexane represents a system where the dominant solvent effect is due to the wave-function confinement. In fact, even for a qualitatively correct prediction of the absorption spectrum, confinement of the solute wave function has to be included in the model. This also explains why the smaller, compact basis sets are able to identify the intense transition, since such basis sets do not allow the wave function to spread out. The consequence of the confinement is visible through the spatial form of the orbitals dominating the lowest excitation in cholesterol (see Fig. 3). The PDE-TDDFT calculations result in a localized $\pi \rightarrow \pi^*$ excitation that is dominated by a HOMO $\rightarrow$ LUMO transition and leads to an intense absorption. On the other hand the lowest excitation obtained through the PE-TDDFT calculations, which is also dominated by a HOMO $\rightarrow$ LUMO transition, is unphysical since the LUMO has escaped into the solvent. The low excitation energy of 5.13 eV indicates a stabilization of the diffuse LUMO in the solvent environment. This can be explained by a too favorable interaction between the electron density of cholesterol and the charge distribution of the solvent. Furthermore, due to the diffuse nature of the LUMO there is virtually no spatial overlap with the HOMO, and consequently the intensity of this transition vanishes.

To verify the results obtained using PDE, the electronic transition properties of cholesterol were evaluated in a full QM calculation of a smaller cluster consisting of a cholesterol molecule with its closest 22 cyclohexane neighbors. Due to the large size of the system (470 atoms), we employed TDHF and mixed basis sets: for cholesterol, we used aug-cc-pVDZ on C and O atoms, and cc-pVDZ on H atoms; for cyclohexane we used 6-31G* on all atoms. The four lowest excitation energies were computed and are presented

TABLE I. Excitation energies (ΔE) and oscillator strengths (f) for the four lowest excitations of cholesterol in gas phase or solvated in 22 cyclohexane molecules described by PDE or included in the QM system (QM cluster). Basis set on cholesterol C,O: aug-cc-pVDZ, H: cc-pVDZ. Basis set on cyclohexane: 6-31G*. Energies are given in eV.

System/Model	State	ΔE	f
Gas phase	1	6.64	0.03
	2	6.97	0.27
	3	7.38	0.10
	4	7.65	0.05
PDE	1	7.09	0.46
	2	7.98	0.07
	3	8.39	0.00
	4	8.67	0.00
QM cluster	1	6.97	0.35
	2	7.59	0.08
	3	7.97	0.00
	4	8.23	0.01

in Table I. The corresponding PDE results, i.e., using the same basis sets, are also given together with the gas phase values. The intense excitation energy calculated with the PDE model is overestimated by ~ 0.1 eV compared to the cluster value, but there is a good qualitative agreement between the PDE and the cluster results. Even though the HF gas phase calculations predict a too large intensity in gas phase compared to RI-CC2 as discussed previously, it is clearly seen from the cluster calculations that the repulsion effects from the solvent molecules—be they quantum-mechanically described or modeled by the repulsion operator—lead to a more intense absorption and less diffuse excited states in solvent compared to in gas phase.

Finally, to include finite temperature effects, the transition properties of 20 snapshots from the MD simulation were evaluated based on PDE-TDDFT calculations with the aim of constructing a spectrum that can, to some extent, be compared with the experimental spectrum (see blue line in Fig. 4). The solvent conformations are sampled, but the intramolecular temperature effects are not taken into account since the geometry of cholesterol was constrained during the simulation. Some discrepancy with respect to experiment is therefore expected. Based on these calculations the average excitation energy for the $\pi \rightarrow \pi^*$ transition is predicted to be 6.81 eV with a standard deviation (σ) of 0.03 eV. The $\pi \rightarrow \pi^*$ excitation energy for each snapshot with its corresponding oscillator strength was convoluted using a

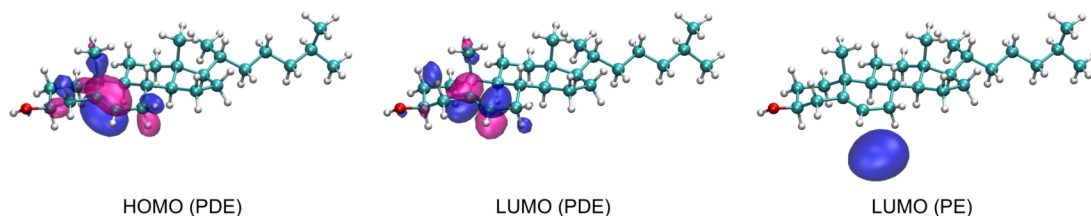


FIG. 3. Isodensity surfaces of the orbitals dominating the lowest electronic excitation in cholesterol. Left: the HOMO obtained from the PDE-TDDFT calculation (the HOMO from PE-TDDFT has a similar shape). Middle: the LUMO obtained from the PDE-TDDFT calculation. Right: the LUMO obtained from the PE-TDDFT calculation. An isovalue of 0.04 a.u. has been used and the figures were produced with VMD.²¹ Solvent molecules are omitted for clarity.

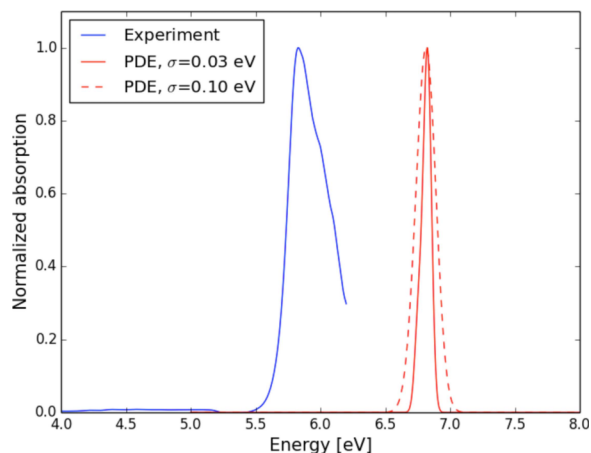


FIG. 4. Normalized experimental absorption spectrum of 2.59 mM cholesterol solvated in cyclohexane (blue) and absorption spectra calculated using PDE-CAM-B3LYP/aug-cc-pVDZ with $\sigma = 0.03$ eV (red full line) and $\sigma = 0.10$ eV (red dashed line).

Gaussian function, and the functions for all snapshots were subsequently added to construct the final spectrum. In all snapshots this transition clearly dominates the absorption, as judged by its oscillator strength. The normalized spectrum can be seen in Figure 4 using $\sigma = 0.03$ eV as obtained from the calculations (red full line) and for $\sigma = 0.10$ eV (red dashed line) to estimate the broadening due to the dynamical effects on the cholesterol geometry and the broadening related to the excited-state lifetime. The blue-shift of ~ 1 eV compared to the experiment can partly be explained by the use of the CAM-B3LYP functional, which has previously been shown to overestimate the excitation energy of $\pi \rightarrow \pi^*$ transitions.^{30,35} Another reason for this deviation between the position of maximum absorption in the experiment and from the calculations may be due to missing effects of solute-solvent dispersion. Inclusion of solute-solvent dispersion would lead to a red-shift in the predicted excitation energies and would thus bring the simulated and experimental data in better agreement. However, the main point is that solute confinement gives rise to a single and well-characterized peak in perfect qualitative agreement with experiment.

In summary, we have here seen a clear example of how the well-established strategy for improving the quality of gas phase calculations within an electronic structure method—increasing the basis set size—can fail when it comes to describing molecules in a condensed phase based on effective Hamiltonian strategies. In this case, the lack of modeling of the quantum nature of the environment leads to a violation of the Pauli principle and consequently to unphysical results when a large basis set is used. We have demonstrated a sophisticated way of calculating the absorption spectrum of a large molecular complex, i.e., a cholesterol molecule in a structured environment, that includes explicit solvent effects: electrostatics in the form of frozen densities and polarization of the environment through distributed polarizabilities, non-electrostatic solvent effects addressed by a repulsion operator, dynamical effects from the MD simulation, and (empirical) lifetime broadening through the σ - and γ -parameters used in resonance-divergent and resonance-convergent TDDFT,

respectively. The scheme presented here is a more robust approach compared to classical embedding, and we therefore expect that it can be used for a wide range of solute-solvent systems where the dominant nature of the solvent effect is not known beforehand. Importantly, the model can be used to rationalize solvent effects in non-polar solvents where electrostatics may play a minor role. In such cases dispersion might also be of significant importance, however, as shown in this presentation, leaving out non-electrostatic repulsion produces even qualitatively wrong results. We therefore point to the latter solvent effect as the most important of these two, especially in apolar solvents and in cases where the isolated molecule has diffuse excited states.

SUPPLEMENTARY MATERIAL

See [supplementary material](#) for further computational and experimental details, reference gas phase calculations, and a comparison of different basis sets for the outer region in PDE.

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